Supplementary Material for CEARF-N: Validation-Gated Evidence Admission for Sparse Session Recommendation

Pham Quoc Cuong[1,2], Nguyen Ngoc Minh[1,2], Le Quang Minh[1,2], and **Nguyen Hoai Nguyet An**[1,2]

[1] University of Information Technology, Ho Chi Minh City, Vietnam
[2] Vietnam National University Ho Chi Minh City, Vietnam
{cuongpq.20,minhnn.20,minhlq.20,annhv.20}@grad.uit.edu.vn

Abstract. This document supplements the eight-page short paper. It specifies the complete scoring functions, candidate families, nested selection algorithm, per-domain neural configurations, per-seed router decisions, external-expert subsets, query-regime and shortcut diagnostics, teacher interactions, leakage controls, and reproduction artifacts. Its additional stratified results are descriptive audits of already locked predictions; the purpose is to make every admission, rejection, and stated boundary independently auditable.

The frozen source and artifact repository is github.com/pzcuong/product-recommendation-system.

1 Notation and Prediction Task

A query is a session prefix $q = (i_1, \ldots, i_L)$ with one held-out next item y_q. Training item frequency is f_x, the catalogue is $\mathcal{I}$, and $\pi_r(q)$ is the ranked list emitted by evidence source r. Evaluation ranks the full catalogue. Seen items are masked on the Amazon protocols and retained on Diginetica, whose protocol permits repeat consumption.

The sole selection objective is

$$U_V(c) = \frac{1}{2}\text{Recall@6}_V(c) + \frac{1}{2}\text{Recall@20}_V(c).$$

At most 5,000 validation queries are ordered deterministically by a BLAKE2b hash. For validation memory construction, the held-out target and its incoming transition are removed from the index. The final index and neural model are rebuilt on all training sessions only after configuration lock.

2 Exact CEARF Memory

2.1 Directed transition evidence

For source item i and candidate x, let $c(i, x)$ be the training count of $i \to x$ within a four-step window, weighted by inverse distance during index construction. For

Table 1. Full-catalogue datasets used by the main paper.

Dataset	Items	Train events	Train seq.	Test queries	Items ≤ 5
Amazon Video Games	25,613	625,062	94,762	94,762	27.91%
Amazon Baby Products	36,014	939,529	150,777	150,777	27.89%
Diginetica	43,098	906,140	186,670	60,858	19.28%

the eight most recent context items,

$$T_q(x) = \sum_{a=0}^{\min(7,L-1)} \frac{\exp(-0.35a)\log(1+c(i_{L-a},x))}{\sqrt{1+f_x}}.$$

The denominator prevents globally frequent items from dominating local transitions. The 200 largest outgoing transition counts per source item are retained before query-time aggregation.

2.2 Neighbour-session evidence

Let $\mathrm{df}(i)$ be the number of indexed sessions containing i, N_S the number of sessions, and $\mathrm{idf}(i) = \log(1 + N_S/\max(1, \mathrm{df}(i)))$. A training session s receives overlap

$$o(s,q) = \sum_{a=0}^{\min(7,L-1)} \mathrm{idf}(i_{L-a})e^{-0.20a}\mathrm{I}[i_{L-a} \in s].$$

The 120 largest-overlap sessions are retained. Overlap is divided by the square root of the session's distinct-item count. A candidate after the most recent context match receives direction weight 1; an earlier candidate receives 0.25. Distance from that match is discounted by inverse square root, and votes are summed across sessions.

2.3 Rank-space memory fusion

Transition, session, and popularity scores are independently converted to top-120 lists. For profile $\mathbf{w} = (w_T, w_S, w_P)$,

$$\widetilde{M}_q(x) = \sum_{r\in\{T,S,P\}} \frac{w_r}{20 + \mathrm{rank}_{\pi_r(q)}(x)}.$$

If $v_q(x) \geq 2$ active components retrieve x, the implementation applies the declared consensus multiplier $M_q(x) = \widetilde{M}_q(x)[1 + 0.12(v_q(x) - 1)]$; otherwise $M_q(x) = \widetilde{M}_q(x)$. The multiplier ablation changes reported Recall@20 by .00000. It is a fixed a-priori scoring constant outside the admission family, retained for exact reproduction and not presented as a rejected optional stage.

Table 2. Declared CEARF memory profiles. Each zero is a component-off state. P, T+P, and S+P are omitted because popularity was scoped a priori as fallback evidence. The family covers four of seven non-empty presence patterns; short and long sessions select independently.

Profile	w_T	w_S	w_P
Transition	1.00	0.00	0.00
Session	0.00	1.00	0.00
Balanced	0.50	0.40	0.10
Transition–session	0.60	0.40	0.00
Session–transition	0.35	0.60	0.05
Short-safe	0.65	0.20	0.15

3 Exact PASGR Residual

PASGR means *Prototype-Aligned Semantic Graph Retrieval*. It is the optional "N" path of CEARF-N, not the expansion of CEARF. PASGR labels the declared candidate family; selected cells may reject prototype transport and do so in all three domains.

3.1 Item initialization

Teacher vectors are reduced or padded to 64 dimensions and normalized. Prototypes are fitted on the better-observed half of items. With prototype membership vector p_i, centre matrix C, and trust $\rho_i = \sqrt{f_i/(f_i + 10)}$, optional transport gives

$$a_i = \rho_i e_i + (1 - \rho_i) p_i C.$$

An undirected adjacency view is built from consecutive training items. If g_i is the normalized mean of neighbour vectors, optional graph aggregation initializes the trainable item vector as $h_i = \mathrm{norm}(a_i + \lambda_g g_i)$.

3.2 Sequence encoder and objective

A one-layer GRU encodes the last 50 items. Its final state passes through layer normalization and a linear projection to obtain normalized query vector z_q. Candidate logits are a learned-temperature dot product with normalized item vectors. Each update contains the target, 32 prototype/graph hard negatives, and random fill negatives. The training loss is

$$\mathcal{L}_N = \mathcal{L}_{\text{rank}} + \lambda_{\text{ib}} \mathcal{L}_{\text{inbatch}} + \lambda_{\text{sem}} [1 - \cos(z_q, \bar{e}_q)],$$

where $\bar{e}_q$ is the exponentially recency-weighted frozen-teacher context view. Repeated in-batch targets are treated as multiple positives.

The complete grid is the Cartesian product $\lambda_g \in \{0, .35\}$, PT$\in \{\text{off}, \text{on}\}$, $\lambda_{\text{sem}} \in \{0, .15\}$, and $\lambda_{\text{ib}} \in \{0, .10\}$. Exact ties prefer fewer active mechanisms.

Table 3. The 16-cell PASGR family and selected cells. "PT" is prototype transport. Selection uses seed 42 validation only; the cell is then locked for three matched-seed evaluation.

Domain	λ_g	PT	λ_{sem}	λ_{ib}	U_V
Video Games	.35	off	.15	.10	.1369
Baby Products	0	off	.15	0	.0488
Diginetica	.35	off	0	.10	.4554

Table 4. Diagnostic cross-seed PASGR stability. Values are mean validation utility over seeds 42/123/456 for the locked seed-42 cell (Sel.), its runner-up (RU), and the best prototype-on cell (PT). The diagnostic does not reselect the canonical system.

Domain	Sel.	RU	PT	Mean winner (margin over Sel.)
Video Games	.13640	.13617	.13347	Sel.
Baby Products	.04663	.04740	.04677	RU (+.00077)
Diginetica	.45603	.45620	(same RU/PT)	RU/PT (+.00017)

4 Adaptive Fusion and Routers

For memory rank π_M and neural rank π_N,

$$F_q(x) = \frac{1-\beta_q}{20+\text{rank}_{\pi_M}(x)} + \frac{\beta_q}{20+\text{rank}_{\pi_N}(x)}.$$

The 21-point grid contains $\beta = 0$ and $\beta = 1$. Three router families compete.

1. **Regime:** one β for $L \leq 2$ and one for $L > 2$.
2. **Bucketed:** 12 cells from length (short/mid/long), last-item popularity (head/tail), and top-5 cross-memory agreement (low/high). Empty cells fall back to regime β.
3. **Continuous:** ridge regression from length, log length, log last-item frequency, tail flag, length bucket, agreement, and transition/session top-5 rank-occupancy mass. Its target is the smallest grid β placing y_q in the fused top 20.

The two occupancy-mass features depend only on the number of valid entries in each top-5 list. They are constant on 100% of both Amazon test sets and take their modal value on 98.94%/99.84% of Diginetica transition/session lists. They must not be interpreted as transition certainty. Effective branching and top-one transition share below are post-hoc train-graph diagnostics, not router inputs.

Router parameters are fitted on a deterministic 80% of validation queries. The family is chosen on the disjoint 20% using U_V, with ties preferring regime over bucketed over continuous. The chosen family is refitted on all validation queries before test inference.

Using regime as the router null, the admission/retention diagnostics are:

Table 5. Router family selected by validation for each neural seed.

Domain	Seed 42	Seed 123	Seed 456
Video Games	Bucketed	Regime	Regime
Baby Products	Continuous	Bucketed	Bucketed
Diginetica	Regime	Regime	Bucketed

Table 6. Per-seed router admission margin A, test retention R, and reversal Z. Values use the common $U = .5R@6 + .5R@20$; regime selections have $A = R = 0$ by definition.

Domain	Seed	Selected	A	R	Z
Video Games	42	Bucketed	.00200	-.00070	1
	123	Regime	.00000	.00000	0
	456	Regime	.00000	.00000	0
Baby Products	42	Continuous	.00400	-.00634	1
	123	Bucketed	.00050	.00004	0
	456	Bucketed	.00150	-.00153	1
Diginetica	42	Regime	.00000	.00000	0
	123	Regime	.00000	.00000	0
	456	Bucketed	.00200	.00030	0

Three of nine decisions reverse on test (Video 1/3, Baby 2/3, Diginetica 0/3). This is why adaptive routing is not presented as a universal accuracy claim.

5 End-to-End Locked Selection Algorithm

1. Declare constants, U_V, candidate families, null states, seeds, and tie-breaks before test evaluation.
2. Remove validation targets and incoming transitions from the tuning index; construct CEARF component lists.
3. Select short/long CEARF profiles from the six-profile family.
4. For each of 16 PASGR cells, train on tuning sessions, produce full-catalogue validation ranks, and select its best β. Lock the cell maximizing U_V.
5. For each evaluation seed, train the locked PASGR cell on tuning data. Split validation 80/20; fit the three router families on 80% and select the family on 20%.
6. Refit the selected router family on full validation. Rebuild CEARF and retrain PASGR on all training sessions using the locked decisions.
7. Score the test set once. Persist query order, rankings, configuration, fingerprints, and paired outcomes.
8. Separately form the hard expert family from CEARF-N, STAN, V-SKNN, NARM, and all 15 non-empty subsets. Select with the same U_V; the four singletons can reject ensembling, and no member, including CEARF-N, is protected. The empty set is inadmissible because a ranking is required.

Table 7. Structural difference between a common tuned hybrid and the declared CEARF-N contract. This comparison concerns the search object, not the mere presence of a validation split.

Audit question	Common tuned hybrid	CEARF-N
Can a proposed block vanish?	Often no; tune its strength conditional on inclusion	Explicit absent candidate at the corresponding stage
Are pure endpoints legal?	Full hybrid is privileged	$\beta = 0$ and $\beta = 1$ compete with every mixture
Can adaptivity vanish?	Router class fixed by design	Regime, bucketed, and continuous families compete
What happens to rejection?	Often omitted or shown only as test ablation	Recorded as a method output
Can the proposed method lose?	Baseline is outside the search	Hard family may reject CEARF-N itself
What does a test reversal mean?	Often prompts redesign	Reported validation-to-test selection gap

6 Why This Is Not Ordinary Validation

The contract is finite and scoped. It does not claim that all possible constants or architectures are searched. It requires only that every *declared optional mechanism* has a comparable null state and that rejections remain visible.

For an admitted stage j, we record $A_j = U_V(c_j^\star) - U_V(c_j^0)$ (admission margin), $R_j = U_T(c_j^\star) - U_T(c_j^0)$ (test retention), and $Z_j = \mathrm{I}[A_j > 0 \wedge R_j < 0]$ (reversal). A rejected null has no post-selection superiority claim. These diagnostics turn the declared family into an auditable object rather than renaming an ordinary tuning grid.

7 External-Expert Decisions

Let C, S, V, and N denote CEARF-N, STAN, V-SKNN, and NARM. Reselection below uses exactly the core $U_V = .5R@6 + .5R@20$, not the earlier exploratory $R@10/R@20$ objective. Candidate fusion is equal-weight RRF with $k = 20$. Every per-seed JSON contains 15 candidates, including C, S, V, and N singletons.

Table 8. Validation-selected hard-expert subsets by seed.

Domain	Seed 42	Seed 123	Seed 456
Video Games	C+S+N	C+N	C+S+N
Baby Products	C+S+V	C+S	C+S+V
Diginetica	C+V+N	C+S+N	C+S+N

Table 9. External-gate diagnostics against the per-seed validation-selected singleton. A and R are mean utility margins; Δ_{20} has a query-clustered 95% interval over matched seeds.

Domain	Mean A	Mean R	Δ_{20} [95% CI]
Video Games	.00493	.00569	.00658 [.00558,.00757]
Baby Products	.00363	.00354	.00518 [.00467,.00569]
Diginetica	.00507	.00638	.00384 [.00212,.00556]

This table is the direct defense against a protected-method interpretation. NARM is rejected on every Baby seed; V-SKNN is rejected on every Video seed; and Diginetica always retains NARM. All four singletons were legal, but none was selected. The paired intervals exclude zero and no external-gate admission reverses on test; the stable object is the admission procedure, not a universal fixed subset.

7.1 Aggregate validation-to-test transfer

The following audit is recomputed directly from the locked router and external gate records under the common utility; it is descriptive because seeds within a domain are not independent datasets.

Table 10. Transfer of positive validation admissions. $\bar{A}$ is the mean validation admission margin and $\bar{R}$ the corresponding test margin.

Gate	Admit/eligible	Keep	Reverse	$\bar{A}$	$\bar{R}$
Adaptive router	5/9	2	3	.00200	-.00164
External experts	9/9	9	0	.00454	.00520

Across the two declared gate types, 11/14 positive admissions retain a non-negative test margin. The contrast is the substantive result: the external gate transfers consistently, whereas small router-selection margins are fragile. Test outcomes are never used to revise a selected family. Across the three seeds, modal admit/reject agreement is 2/3, 3/3, and 2/3 for the Video, Baby, and Diginetica routers, compared with 3/3 for every external gate. Exact selected router-family and expert-subset agreement is 2/3 in every domain. Thus admission stability and configuration stability are distinct.

Scope and gate comparability. Transfer is audited only where locked predictions exist for both the selected state and its null: adaptive router versus regime, and expert subset versus the validation-best singleton. We do not infer test retention for individual memory profiles, PASGR switches, or fusion weights.

Router families are fitted on 4,000 validation queries and selected on 1,000; the 15 expert subsets use 5,000 validation queries. Their utility-margin resolutions are therefore .0005 and .0001. The gates also differ in family size and null definition, so their reversal rates diagnose the realized procedures rather than identify a causal advantage of one gate type.

Selection and membership stability. The persistent expert cores across seeds are C+N, C+S, and C+N on Video, Baby, and Diginetica; intersection-over-union ratios are .667, .667, and .500. Across all nine decisions, C/S/V/N are retained 9/7/3/6 times. Despite changing membership, selected-gate Recall@20 is .153423±.000673, .058815±.000104, and .537897±.002984. Configuration membership can therefore vary while the reported system metric remains stable.

Selector lineage. The slim canonical artifacts retain target ranks rather than complete top-20 lists, so the external audit reconstructs CEARF-N lists by refitting the same pipeline, configuration, and seed. On Diginetica seed 42, this realization has Recall@20 .49757 rather than the canonical .51913. Validation selection and test evaluation consistently use the refitted realization, so no test reselection occurs, but it is not a bit-identical reuse of the canonical prediction and is excluded from canonical output-membership claims.

8 Evidence-Necessity and Shortcut Diagnostics

This audit uses only target-rank vectors from the already locked test predictions. It does not tune a threshold, router, or model on test. A rescue is a query for which memory misses the target at 20 and selected fusion hits; damage is the converse. By construction,

$$\Delta\mathrm{Recall@20}_{F-M} = P(\mathrm{rescue}) - P(\mathrm{damage}).$$

Table 11. Constituent endpoints and selected fusion, three-seed means. Each cell reports Recall@6/10/20.

Dataset	Memory only	Neural only	Selected fusion
Video	.06932/.08821/.11901	.06158/.08471/.12571	**.07803/.10368/.14685**
Baby	.02288/.02900/.03879	.02337/.03214/.04873	**.02767/.03668/.05346**
Diginetica	.29174/.37334/.49565	.26398/.33539/.44852	**.30738/.39210/.51681**

Table 12. Query-level evidence decomposition, three-seed means. R and D are rescue and damage rates over all test queries.

Domain	Memory	Neural	Fusion	R	D	R−D
Video Games	.11901	.12571	.14685	.04695	.01911	.02783
Baby Products	.03879	.04873	.05346	.02791	.01324	.01467
Diginetica	.49565	.44852	.51681	.04455	.02339	.02116

Table 13. Disjoint decomposition of fusion rescues, as three-seed mean query rates. New means absent from the union of transition, session, and popularity top-120 lists; Unselected means present only in a memory component excluded by the locked profile; Promote means selected-memory rank 21–120.

Domain	New	Unselected	Promote	Damage	Net
Video Games	.01553	.00327	.02814	.01911	.02783
Baby Products	.01283	.00504	.01004	.01324	.01467
Diginetica	.00079	.00004	.04372	.02339	.02116

New+Unselected+Promote−Damage equals Fusion−Memory. Of all rescues, 33.1%, 46.0%, and 1.8% are New on Video, Baby, and Diginetica. The first two domains therefore contain genuine candidate introduction, whereas the Diginetica residual is overwhelmingly a reranker of retrieved candidates.

Table 14. Net rescue rate by target-popularity stratum. Popularity uses a training-frequency item-rank 20/60/20 head/torso/tail split.

Domain	Head	Torso	Tail
Video Games	.03310	.02339	.00569
Baby Products	.02022	.00225	-.00035
Diginetica	.01228	.03070	.03713

Table 15. Net fusion-minus-memory Recall@20 under router-observable test covariates. Popularity refers to the observed last context item, not the unknown target. A dash denotes an empty stratum.

Signal	Level	Video	Baby	Diginetica
Length	$L \leq 2$	–	–	.01436
	$3 \leq L \leq 7$	.02725	.01504	.02713
	$L > 7$	.02931	.01365	.02556
Last-item popularity	Head	.02627	.01529	.01485
	Tail	.03288	.01220	.04068
Memory agreement	Low	.02874	.01560	.02565
	High	.02710	.01374	.02002

Length-cell counts are 0 / 68,091 / 26,671 (Video), 0 / 110,156 / 40,621 (Baby), and 27,409 / 25,091 / 8,358 (Diginetica). Target-popularity and last-item-popularity results answer different questions: only the latter is observable by the router at inference.

Table 16. Net fusion-minus-memory Recall@20 by validation-frozen transition diagnostics. Branching is effective outgoing branch count; certainty is the largest outgoing probability. Unsupported final items remain separate.

Diagnostic	Stratum	Video	Baby	Diginetica
Branching	Unsupported	.02705	-.00120	.07949
	Low	.03265	.01416	.03514
	Mid	.02489	.01467	.01735
	High	.02552	.01539	.01168
Certainty	Uncertain	.02667	.01512	.01260
	Mid	.02584	.01505	.02067
	Clear	.03079	.01404	.03047

Effective-branch terciles are frozen on validation and applied unchanged to test; zero-outgoing contexts are never folded into “low”. The patterns do not support a universal claim that neural evidence helps more as branching increases. Low cross-memory agreement is the more directionally consistent locator.

Table 17. Raw memory-component target coverage and locked transition-removal audit. Union is coverage by any component, not an achievable ranking. No-T changes only the locked memory path.

Domain	T	S	P	Union	Memory	Fusion	No-T	Δ
Video	.04223	.11901	.03787	.14621	.11901	.14685	.11901	.00000
Baby	.00735	.03879	.02908	.06147	.03879	.05346	.03879	.00000
Diginetica	.40092	.49427	.00930	.55271	.49565	.51681	.49354	-.00210

All Amazon test contexts use the locked long profile $(0, 1, 0)$, so transition weight is inactive throughout. Diginetica activates it on 54.96% of queries; removal changes memory by -.00383 within that subset. This does not remove PASGR graph evidence, refit PASGR, or reselect the router, and hence is a memory-path diagnostic rather than a causal full-system ablation.

The Amazon residual is concentrated on frequent targets, whereas Diginetica shows the opposite gradient. Consequently, a generic "neural helps the long tail" interpretation is falsified by Baby Products. Context-length net gains remain positive in every domain stratum. Splitting memory evidence by top-5 agreement gives low/high net gains of .02874/.02710 on Video, .01560/.01374 on Baby, and .02565/.02002 on Diginetica. Neural admission is therefore somewhat more valuable when the three memory sources fail to agree, but the difference is modest.

The outcome regions further bound the claim. Memory and neural both miss 82.18%, 92.80%, and 42.83% of Video, Baby, and Diginetica queries. Within the neural-only region, selected fusion retains 73.20%, 80.01%, and 52.21% of neural hits. These observations establish complementary top-20 evidence, not causal immunity to benchmark shortcuts: no test intervention removes local transitions or similar sessions.

9 Statistics and Metadata Fairness

Recall@20 comparisons use paired binary outcomes in common query order. Confidence intervals use 20,000 query-clustered bootstrap repetitions. McNemar's exact test uses the two discordant counts. Seed means and standard deviations use the independent seeds 42, 123, and 456. Diginetica contributes one next-item query per retained prefix. Original source-session identifiers are unavailable after preprocessing, so intervals resample prefixes and may underestimate dependence among prefixes from the same source session.

The audit supplies the same ID-only, TF–IDF/SVD, or MiniLM information condition to both CEARF-N and NARM within each comparison, while preserving each system's validation-selected checkpoint/configuration. Teachers initialize item representations offline; no test label or test interaction is used to build them. MiniLM improves CEARF-N but does not overturn matched-teacher NARM, which prevents semantic quality from being credited to fusion.

Table 18. Teacher audit: Recall@20 mean (standard deviation), three seeds. “None” is ID-only.

Dataset	Model	None	TF–IDF/SVD	MiniLM
Video	CEARF-N	.13208 (.00056)	.14685 (.00038)	.14794 (.00114)
	NARM	.13705 (.00081)	.15387 (.00051)	.15392 (.00025)
Baby	CEARF-N	.05055 (.00065)	.05346 (.00397)	.05761 (.00058)
	NARM	.02990 (.00018)	.06628 (.00014)	.06493 (.00005)

Table 19. Architecture gap Recall@20(CEARF-N)−Recall@20(NARM). Positive values favour CEARF-N.

Domain	Teacher	Overall	Head	Torso	Tail
Video	None	-.00497	-.01146	+.00666	+.00532
	TF–IDF	-.00702	-.01653	+.01072	+.00608
	MiniLM	-.00598	-.01027	+.00089	+.00308
Baby	None	+.02065	+.02667	+.00795	+.00216
	TF–IDF	-.01282	-.01901	+.00219	+.00055
	MiniLM	-.00732	-.01171	+.00355	+.00150

Descriptive teacher interaction. For teacher t, let $G_t = R@20(\mathrm{C}, t) - R@20(\mathrm{N}, t)$ and $I_t = [R@20(\mathrm{C}, t) - R@20(\mathrm{C}, 0)] - [R@20(\mathrm{N}, t) - R@20(\mathrm{N}, 0)]$. For TF–IDF/MiniLM, I_t is -.00206/-.00101 on Video and -.03346/-.02797 on Baby. These are system-level differences, not a causal factorial decomposition: the no-metadata CEARF-N condition uses the default PASGR cell and regime router, whereas semantic conditions use teacher-specific validation-selected PASGR cells, routers, and checkpoints. Aggregate matched-teacher NARM wins are head-driven; positive mean torso/tail gaps should likewise not be read as universal seed-level superiority.

Legacy alignment record. The TF–IDF NARM rank writer stored a metric scalar in its `test_fingerprint` field. Its generation functions nevertheless use sorted test identifiers for both prediction and targets, vector lengths match, and aggregate metrics reproduce the recorded results. The audit script accepts only this known legacy format and lists every exception explicitly in `teacher_interaction_audit.json`.

10 Reproducibility Manifest

The main runs used an Apple M2 Pro with 32 GB memory. CEARF inference is single-process CPU code; PASGR uses Metal. Query fingerprints and rank arrays allow all paired tests to be recomputed without retraining.

Table 20. Primary implementation and evidence artifacts relative to `sparse_bench/`.

Artifact	Purpose
`cearf.py`	CEARF index, component scores, profiles, validation split
`pasgr.py`	PASGR initialization, losses, training, full-catalogue inference
`perquery_router.py`	Bucketed and continuous routers
`run_validation_gated_pasgr.py`	16-cell PASGR selection
`run_cearfn_v2.py`	Canonical nested CEARF-N runner
`pasgr_config_per_domain.json`	Locked PASGR choices
`cearfn_v2_nested_results.json`	Per-seed router and test records
`narm_expert_matched_r6r20.json`	Consistent-objective expert audit
`hard_gate_singleton_audit.json`	Paired selected-gate versus validation-best-singleton audit
`validation_test_transfer_audit.json`	Recomputed admission-retention summary
`audit_validation_test_transfer.py`	Transfer-audit reproduction script
`evidence_necessity_audit.json`	Locked rescue/damage and regime audit
`audit_evidence_necessity.py`	Evidence-necessity reproduction script

Offline construction provenance. The canonical training timer is not an inference claim. Per-seed timers wrap PASGR training on the tuning and full sessions, neural prediction on validation and test, router fitting/selection, fusion, metrics, and compressed rank-artifact writing. Domain timers additionally include data loading, CEARF index construction, profile tuning, memory ranks, and query features.

Domain	Setup (min)	Mean/seed (min)	3 seeds (min)
Video Games	13.7	10.9	46.3
Baby Products	19.1	15.8	66.3
Diginetica	6.0	12.1	42.3

Because the canonical runner did not separately instrument post-training serving, these numbers describe experiment construction rather than deployment.

Warm inference benchmark. We separately instrument the deployable path with trained PASGR checkpoints and built CEARF indices. Timed work includes CEARF retrieval to top-120, PASGR full-catalogue prediction to top-120 (batch

Table 21. Reproducibility manifest (continued).

Artifact	Purpose
`shortcut_diagnostics_audit.json`	Memory-component and transition-removal audit
`audit_shortcut_diagnostics.py`	Shortcut-diagnostic reproduction script
`teacher_interaction_audit.json`	Architecture–teacher and popularity audit
`audit_teacher_interaction.py`	Teacher-interaction reproduction script
`runtime_audit.json`	Canonical timer scope and per-domain wall clock
`benchmark_cearfn_inference.py`	Warm full-catalogue inference benchmark
`cearfn_inference_benchmark.json`	Component time, throughput, and amortized time per query
`cearfn_inference_benchmark_pre_reuse.json`	Diagnostic run before exact router-feature rank reuse
`pasgr_stability_audit.json`	Three-candidate cross-seed PASGR stability
`pasgr_stability_artifacts/*.json`	Per-seed candidate utilities and ranks
`CANONICAL_EXPERIMENT_LINEAGE.md`	Submission source-of-truth boundary
`external_baseline_paired_nested*.json`	Paired baseline outcomes

256), router features, selected-router beta assignment, and RRF fusion. It excludes loading, teacher construction, training, index construction, profile/router fitting, warm-up, metrics, and artifact writing. Before timing, each path processes the first 512 test queries once and the Metal queue is synchronized. We report one complete timed pass per domain (60,858–150,777 queries), rather than a mean or median over repeated micro-benchmarks; hence no run-to-run dispersion is claimed. ms/query is total time divided by query count and represents amortized batch throughput, not isolated request latency.

Domain (Q)	Items	Memory	Neural	Router+fuse	Total s	ms/q	q/s
Video (94,762)	25,613	280.40	71.21	10.68	362.29	3.823	261.6
Baby (150,777)	36,014	486.62	121.16	17.49	625.28	4.147	241.1
Diginetica (60,858)	43,098	118.10	29.67	6.63	154.41	2.537	394.1

Memory retrieval accounts for 76.5–77.8% and PASGR for 19.2–19.7% of total time; beta assignment is below 0.05%. The first diagnostic Video run recomputed CEARF rankings while extracting router features and required 620.93 s (6.553 ms/query). Reusing the already emitted transition, session, and popularity lists is rank-identical under `test_cearfn_inference_reuse.py` and reduces this to 362.29 s (3.823 ms/query); the paper reports the non-duplicating implementation.

Canonical lineage. All paper tables use `cearfn_v2_nested_results.json`, the matched NARM checkpoints, and the $R@6/R@20$ reselection in `narm_expert_matched_r6r20.json`. The earlier Diginetica .49028/.48265 continuous-router exploration is excluded; the locked official split yields CEARF-N .51681 and NARM .53406. Per-seed candidate JSON files and selector artifacts preserve the hard-audit query order.

11 Falsification Checklist

The method's claims would be weakened by any of the following observable outcomes: (i) selected fusion failing to exceed both direct endpoints; (ii) the hard expert gate always protecting CEARF-N; (iii) matched metadata eliminating the apparent fusion gain; (iv) router families changing test performance without held-out selection gain; or (v) configuration decisions depending on test labels. The main paper reports both supporting and adverse cases, including the Diginetica loss to ID-only NARM and the repeat-memory validation-to-test reversal.